AI FOR FOUNDATIONAL LEARNING HACKATHON · CHALLENGE 2:
LEARNING-LEVEL VISIBILITY

SUNO सुनो

SUNO listens to the teacher, not the classroom.

A 60-second voice note after the reading period becomes checked, per-child evidence, and a 2-minute Hindi brief the next morning: who needs support with what, today's activity, and whom to listen to.



Live: suno-8qoh.onrender.com · code SUNO-DEMO · video: youtu.be/GG3v7lQ212M

Team Sunvai · Tushar (product, engineering) · Saikiran (architecture, risk)

The teacher already hears every child. By tomorrow, it's gone

76.6%

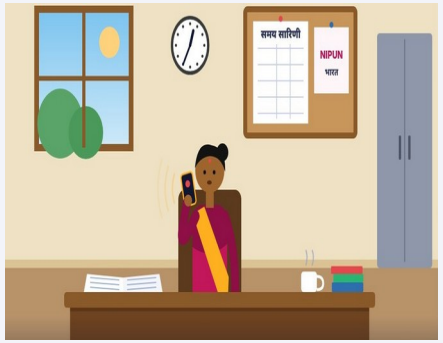
of Class 3 children in government schools can't read a Class 2 text (ASER 2024).

35 children, one teacher, two exams a year: help comes too late.

Assessment apps add a test event, one child at a time. SUNO adds no test: it keeps what she already notices.

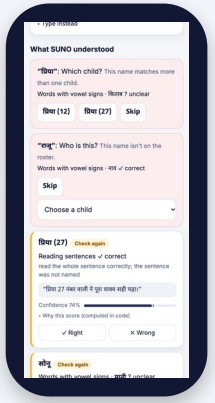
After the reading period

She speaks a 60-second note in her own words: who read what, who got stuck where, who was absent.



SUNO checks

Claude turns it into per-child observations. Code checks every claim against her words and the roster, and computes confidence.



Next morning

A 2-minute Hindi brief: who needs support with what, a 10-minute group activity, and five children to listen to.



Effective AI use and technical strength

Claude where it is strong: messy, code-mixed speech. Code where it must be exact: who, how sure, and what she hears.

AI in layers: from her voice note to the morning brief

Data input

Teacher's voice note
60 s, Hindi or Hinglish, her own words

Class roster
first names + roll numbers (register photo)

This week's lesson
words and sums (textbook photo)

Optional starting levels
ASER-style check

Processing

Sarvam saaras:v4
speech to text; numbered sentences

Claude ① claude-opus-5-5
per-child observations + whole-class remarks; JSON schema; cites sentence IDs

Ladder + planner (code)
Bayesian TaRL levels; picks 5 children to listen to

Claude ②
words the brief from code's decisions

Guardrails (code)

Grounding
every claim cites her real sentences

Names resolved by code
roster, roll numbers; one tap if unsure

Labels masked, opinions held
"weak" / कमज़ोर never evidence

Confidence computed in code
gate 0.75; else "check tomorrow"

Evaluation

Real speech
15 recordings, 3 settings

Pipeline simulation
32 notes, 4 teacher styles

Term simulation
200 terms, plus ASER 2024 classes

Unit + adversarial tests
97 tests, no keys needed

Output

Morning brief
2-minute Hindi voice (Sarvam bulbul:v3), checked against code; fails closed

Class picture
levels, groups, progress

"How SUNO decided"
raw Claude output, every check, the arithmetic

Where each AI call happens, and what it actually does

When	Model	Input → output	What code does with it
Each note	Sarvam saaras:v4	60 s of audio → Hindi transcript, split into numbered sentences	Audio deleted after transcription; the transcript is the only source of evidence
Each note	Claude ① claude-opus-5-5 adaptive thinking, effort high	Transcript + roster + lesson → JSON (observations.v2): child, skill, item, right/wrong, sentence IDs	Checks grounding, resolves the child, masks labels, computes confidence; below 0.75 → "check tomorrow"
Each morning	Claude ② claude-opus-5-5 effort medium	Code's decisions (levels, groups, list) → warm 2-minute Hindi script (brief.v3 schema)	Rejects any child, skill or plan code didn't decide; fails closed to code's own text
Each morning	Sarvam bulbul:v3	Hindi script → MP3 voice (32 kbps)	Cached on the phone for offline listening
Setup only	Claude vision	Register photo → first names; textbook page → words by skill	Keeps first names only; the teacher checks the list; the photo is not stored

Claude API features used: structured outputs (JSON schema), adaptive thinking with explicit effort, prompt caching (95% of note input served from cache), server-side fallback, image input. Versioned prompts in prompts/, schemas in schemas/.

Guardrails: wrong AI output never reaches a child's record

Grounded in her words

Every claim must cite the teacher's real sentence IDs, and its words must appear there.

Code names the child

Phonetic match across Devanagari and Latin; roll numbers; two Priyas → one tap, never a guess.

Labels masked

"Weak", कमज़ोर and similar words (Hindi and English list) are masked and never repeated.

Opinions aren't evidence

"Sonu is weak" becomes "listen to Sonu tomorrow", not a record.

Confidence from code

Computed from measured signals; Claude never sets it. Below 0.75 → "check tomorrow".

Rule of three

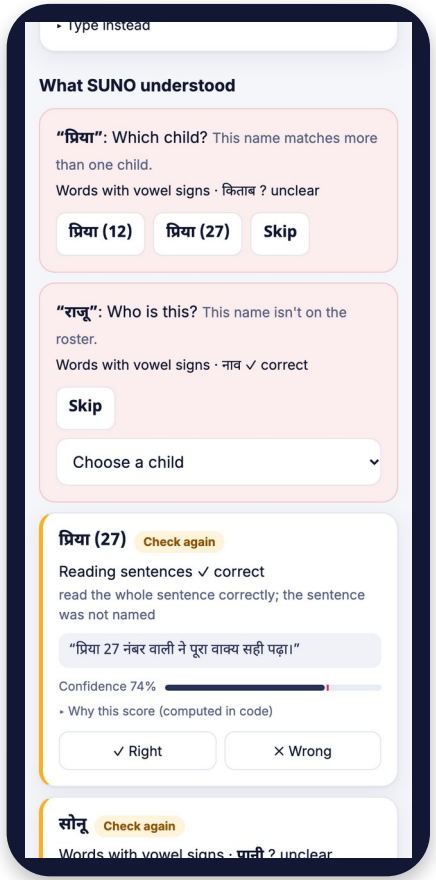
A child is named "needs support" only after 3 observations on 2 days.

Brief checked, fails closed

Claude's brief may only say what code decided, or code's own text is used.

The teacher decides

Right / wrong taps on every record; "How SUNO decided" shows the raw output and every check.



In the app: "which Priya?" is one tap; a name not on the roster is asked, not guessed.

It works on a real voice, on an ordinary phone, in classroom noise

0 / 56

evidence records about the wrong child

0 / 21

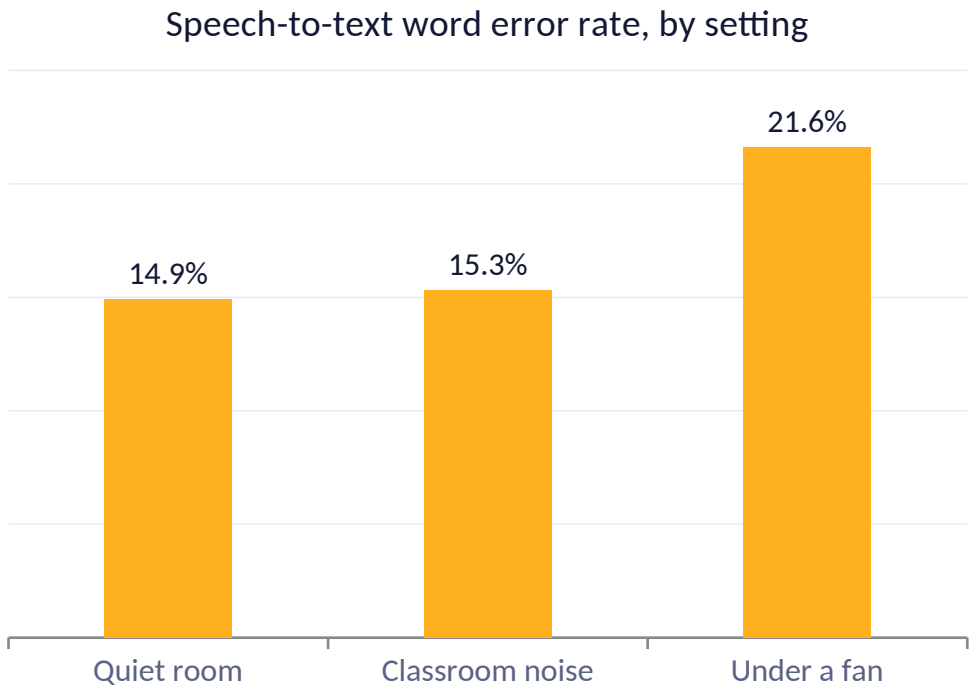
opinions, labels or vague remarks that became evidence

98%

of clear observations captured (56 of 57)

15 / 15

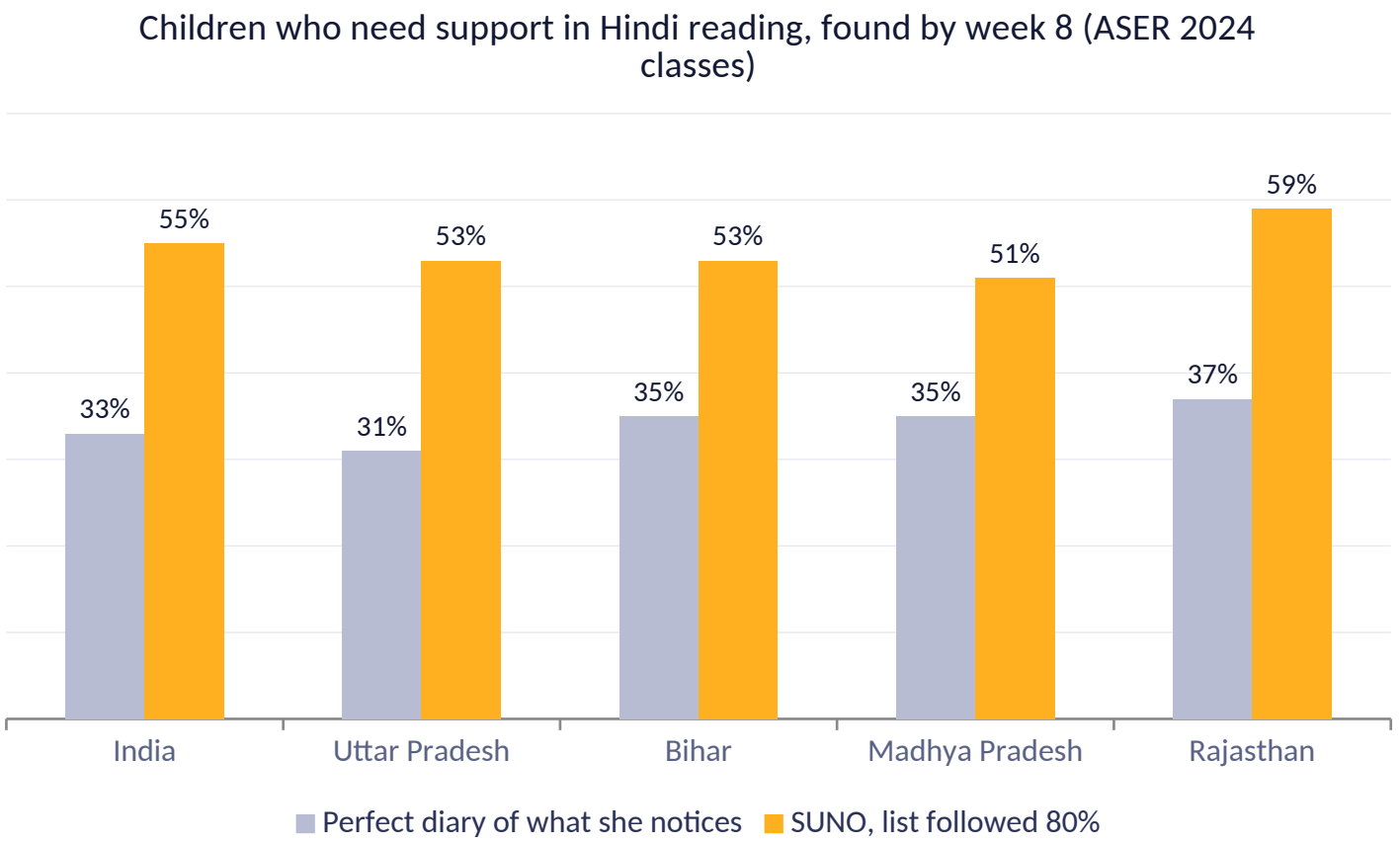
absences recorded, and unclear names resolved by one tap



Despite 15–22% transcription errors, no evidence landed on the wrong child: code resolves names, and a specific word or sum is required.

Real-speech test: 15 recordings of 5 scripted notes by one team member (an adult man) in three settings; a fictional class, so ground truth is known. Not yet a teacher. Source: eval/results/real_notes.md

Simulated terms: the listening list finds the children a diary misses



93% vs 41%

of the quietest third heard by week 2

36% vs 4%

of quiet children's gaps found by week 8

≈ 99%

of "needs support" flags are right, in every state

SIMULATED children and teachers, SUNO's real tracker and planner, error rates from the pipeline tests (floored at 3/n). Chart: reading levels and attendance from ASER 2024's published state tables (eval/results/sim_aser.md). Stat cards: 200 default terms (eval/results/sim_term.md). The first planner made detection worse; the simulation changed the design.

Built for low connectivity, low data and low-cost phones

Works offline

- Recording a note: saved on the phone, sent automatically when the network returns
- The last brief and its audio
- The class picture and child pages as last loaded
- The app itself (installable PWA, no app store)

Needs a network

- Sending a note: about 100–250 KB a minute
- Fetching the next brief: a few KB of text plus ~0.5 MB of audio (32 kbps MP3)
- Setup photos: shrunk to ~1600 px on the phone before upload

Degrades gracefully

- No signal: the note waits in an offline queue, nothing is lost
- A skipped day breaks nothing
- If Claude's brief fails a check, code's own text is used
- Server asleep (free hosting): first visit takes ~1 minute

Try the offline test

1. Turn the phone's network off
2. Record a note: it is saved on the phone
3. Turn the network on: it sends itself, and the result appears

Any Android phone with Chrome (₹8–10k class). About a minute of teacher time a day. ₹17.6 per teacher per school day, ≈ ₹117 per child per year (from logged usage, eval/results/cost.md).

User inputs and testing

Two mentor calls, organiser guidance, a feasibility test and a real-speech test each changed what we built.

What we heard, and what we changed

Source	What they said or showed	What we changed
Technical mentor, 20 Oct	"Test whether each answer maps to the right child before building any app." Recording all day heats and drains a ₹8–10k phone.	Built feasibility test F1 on real child voices: voice matching cannot name the child. Dropped classroom recording; SUNO now listens to the teacher.
Real-speech test, 27 Oct	3 of 21 vague remarks became evidence; 3 of 15 absences lost ("रोल तेरह" heard as "रोल तेरा").	Evidence now needs a specific word or sum; spoken roll numbers parsed; one-tap question for unclear absences. Re-run: 0 of 21, 15 of 15.
Education mentor, 28 Oct Rebecca Swittens, Madhi Foundation	A note after each of ~6 classes is too much.	One note a day after the reading period, plus a daily reminder in her own calendar.
	Teachers speak at class level ("most struggled with matras").	Whole-class remarks become a code-checked "reteach or move on" that opens the brief.
	Levels need a baseline like the ASER tool; show progress over time.	Optional ASER-style starting levels; each child's level week by week.
	Teachers skip children; use TaRL-style nudges. Align to the textbook.	After each note, a one-tap nudge for listed children she didn't mention. Textbook page photo → this week's words.
Organisers, mid-Oct	Real users and settings score higher; show mentor input and what it changed.	This change log (docs/decisions.md); teacher sessions requested for the finale.

Full log with dates and where each change shows in the code: docs/decisions.md. Language per class (e.g. Tamil), also from the education mentor, is on the roadmap.

How the design changed, and what we have not proven yet

1. **Ideate**
   A phone records the reading lesson
2. **F1 test**
   Voice matching can't name the child
3. **Pivot**
   Listen to the teacher, not the classroom
4. **Simulation**
   First planner made detection worse → ladder + two-item check
5. **Real speech**
   Vague remarks slipped in → specific-word rule
6. **Mentor**
   One note a day, class-level remarks, baseline, nudges, textbook

Tested

- One real speaker, 15 recordings, three settings
- 32 simulated notes, 4 teacher styles, real speech-to-text: 0 of 101 records on the wrong child, 0 of 42 opinions became evidence
- 200 simulated terms, and 200 per state in ASER 2024-calibrated classes
- 97 tests, including adversarial cases

Not proven yet, and our plan

- No real teacher has used SUNO in her class yet (sessions requested through the organisers)
- Most teachers are women; our real-speech speaker is a man
- A level from SUNO is not yet compared with an ASER check on the same children: that is the November pilot

Future roadmap

From a validated pilot to a capability inside the apps teachers already use.

Three phases: validate, reach a block, live inside state systems

Nov 2026

Classroom pilot + validation

- 3–5 Grade 1–3 teachers, Hindi-medium government schools, two weeks
- Pre-registered check: SUNO's level vs a blind ASER one-to-one test on the same 20–30 children
- If agreement within one level < 80%, raise the naming threshold
- Measure seconds per note, notes per week, briefs heard

Early 2027

One block, the channel teachers use

- WhatsApp: send the note as a voice note, get the brief back as one
- Block dashboard: counts of children by level per school; no names, no teacher field, no ranking
- State-owned speech: Bhashini or AI4Bharat IndicConformer (one module to swap)
- Seed levels from NIPUN Lakshya or school ORF results

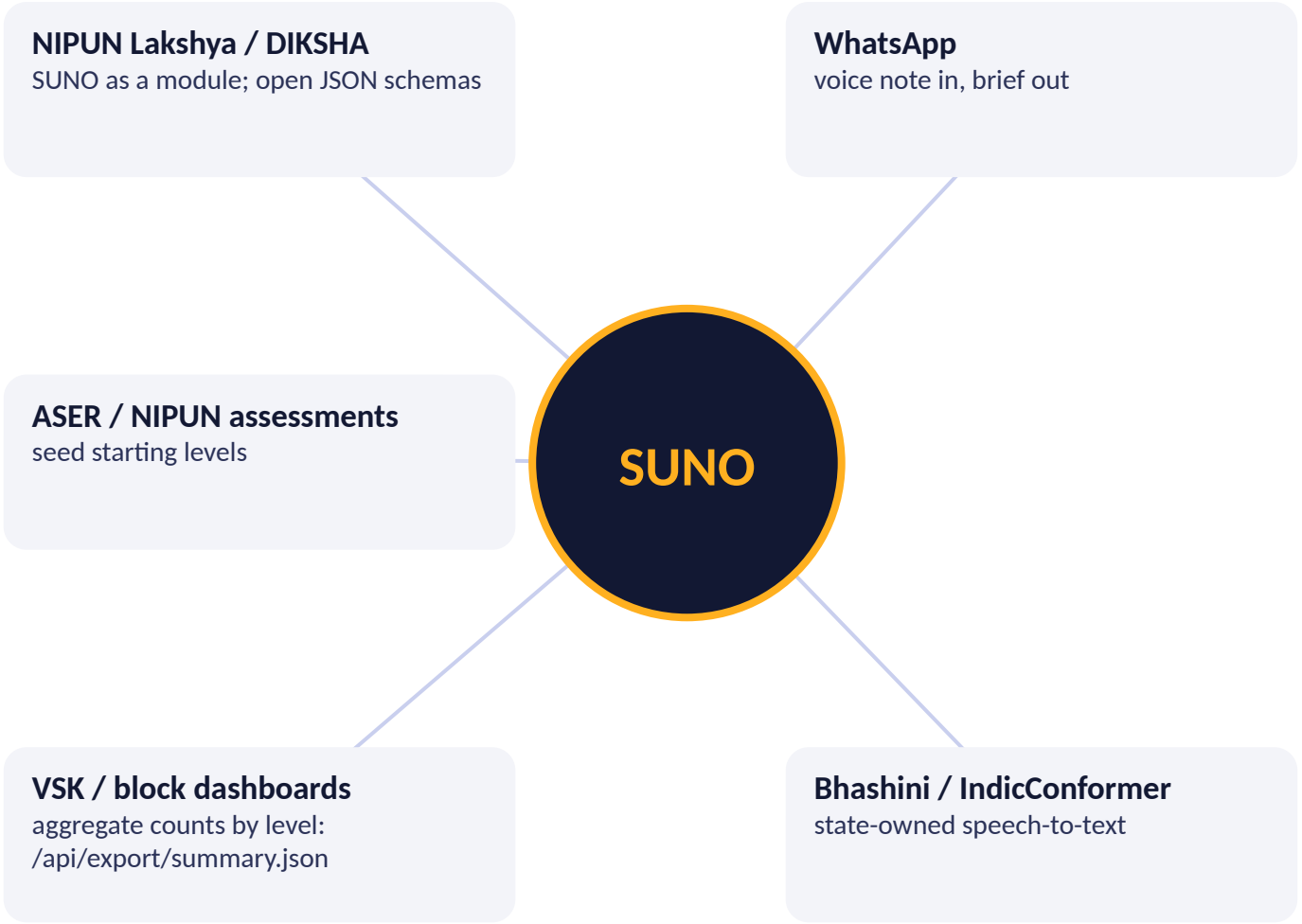
2027–28

Inside state systems

- A SUNO module inside NIPUN Lakshya or DIKSHA, not another app
- More languages: Marathi, Tamil, Odia, each with its own ladder and banned-word list
- The state teacher guide's activities, retrieved by skill
- Coaching only with the teacher's consent

In every phase: no child's voice recorded, no teacher score, first names only, aggregates upward, and the teacher decides.

Integration into existing systems, and who pays



≈ ₹117

per child per year today (₹17.6 per teacher-day, from logged usage). A leaner setup (overnight batch, lower-effort brief) is estimated at about half.

- Who pays: the state, from the Samagra Shiksha FLN budget for teacher support and assessment
- Who owns the data: the school or state; SUNO is the processor (DPDP Act 2023, Rules 2025)
- Never: names upward, teacher scores, teacher rankings

Try SUNO

The teacher already hears every child. SUNO remembers.

Live app suno-8qoh.onrender.com · code SUNO-DEMO (a fresh synthetic class per sign-in), or create your own class

Sample inputs 4 Hindi voice notes (sample notes on the Note tab) + a register photo and a textbook page (☰ menu)

Offline test Turn the network off, record a note, turn it back on: it sends itself

Demo video youtu.be/GG3v7lQ212M (3 minutes)

Source README: setup, local run, .env.example, every AI service and where its key goes

